

General Mathematics-MTH100

Solution Vu360



Here are the solutions:

Question 1: Set Theory Proof

To prove: $(A \cap B)' = A' \cup B'$

1. Start with $(A \cap B)'$.
2. Apply De Morgan's Law: $(A \cap B)' = A' \cup B'$.
3. Thus, $(A \cap B)' = A' \cup B'$ is proven.

Question 2: Composite Function

Given: $f(x) = x^2, g(x) = x + 1$.

1. Find $(f \circ g)(x) = f(g(x)) = f(x + 1) = (x + 1)^2$.
2. Domain of $f \circ g$: All real numbers.

Question 3: Inverse Function

Given: $f(x) = 2x + 5$.

1. Find inverse $f^{-1}(x)$: $x = 2f^{-1}(x) + 5$.
2. Solve for $f^{-1}(x)$: $f^{-1}(x) = \frac{x-5}{2}$.
3. Domain of f and f^{-1} : All real numbers.
4. Verify $f(f^{-1}(x)) = x$: $f\left(\frac{x-5}{2}\right) = 2\left(\frac{x-5}{2}\right) + 5 = x$.

Question 4: System of Equations

System:

1. $2x - y + z = 4$
2. $x + 3y - 2z = 5$
3. $-x + 2y + 3z = 2$

Solution Method: Substitution or Elimination.

Using Elimination:

1. Add Eq. 1 and Eq. 3: $x + y + 4z = 6$.
2. Subtract Eq. 2 from new Eq.: $-2y + 6z = 1$.
3. Solve for y : $y = \frac{6z-1}{2}$.
4. Substitute y into Eq. 2: $x + 3\left(\frac{6z-1}{2}\right) - 2z = 5$.
5. Solve for x : $x = \frac{10-15z}{3}$.
6. Substitute x, y into Eq. 1 for z .

Resulting values of x, y and z depend on correct substitution and elimination steps.